CARLA Simulation Evaluation Report

Project: Vulnerable Road User (VRU) Safety Enhancement via Infrastructure-to-Vehicle (I2V) Communications

Environment: CARLA Autonomous Driving Simulator (Synchronous Mode, $\Delta t = 0.04s$)

1. Executive Summary

This evaluation compares the safety performance of an autonomous **Ego Vehicle** across two operating conditions when encountering pedestrian hazards at an intersection:

1. **Baseline Mode (Unassisted):** Relies strictly on the vehicle's onboard front-facing semantic segmentation camera.
2. **V2X-Assisted Mode:** Fuses onboard perception with real-time early warnings transmitted from an intelligent **Roadside Unit (RSU)** infrastructure camera.

The testing evaluates how early infrastructure line-of-sight offsets local occlusion or high-speed blind spots to mitigate collisions with VRUs.

2. System Architecture & Methodology

2.1 Perception Channels & Pedestrian Detection Logic

Both detection systems rely on **Semantic Segmentation Cameras** rather than standard RGB networks. This allows for mathematically precise object extraction by mapping environmental assets directly to specific color-coded ID tags.

[Raw Camera Stream] ---> [Isolate Red Channel (Labels)] ---> [Count Pixels == 4] ---> [Threshold Check]

- **The Target Class (Pedestrians):** In CARLA's semantic segmentation engine, pixel tags are encoded directly into the **Red channel** of the BGRA image array. The value 4 corresponds explicitly to Walkers (Pedestrians).
- **Onboard Ego Detection Pipeline (semantic_callback_ego):**
  - **Array Parsing:** Reshapes the raw buffer data into an $(H, W, 4)$ array.
  - **Filtering:** Targets the tag channel (labels = img_array[:, :, 2]).
  - **Threshold:** Triggers a detection state if the total sum of pedestrian pixels exceeds **150 pixels**.
  - **Ego Visualization:** Multiplies the raw color tags by 10 (img_array[:, :, :3] * 10) to turn dark semantic tags into bright, visible color segments on screen.

- **Infrastructure RSU Detection Pipeline (semantic_callback_rsu):**
  - **Vantage Sensitivity:** Because the RSU is mounted high above the intersection, pedestrians appear smaller in its field of view. To compensate, its detection threshold is reduced to **40 pixels**.
  - **V2X Message Broadcasting:** Upon triggering, it serializes a payload containing a timestamp and the estimated geographic location of the hazard:

$$\text{Estimated Location} = (x: 130.0, y: 141.0, z: 2.5)$$

2.2 Control & Braking Policies

```
       [ Hazard Detected? ]
                |
      +----------+----------+
      | YES              | NO
      v                  v
 [ Brake Policy ]     [ Autopilot Cruise ]
 - V2X: Smooth (0.5)    - Keep Active
 - Local: Slam (1.0)
```

- **Baseline Emergency Braking:** Enforces a maximum braking force (brake=1.0) upon immediate local visual confirmation.
- **V2X Early Braking:** Enforces a predictive, smoother deceleration curve (brake=0.5) based on network broadcast frames, prioritizing passenger comfort and vehicle stability.

3. Experimental Setup & Scenarios

The evaluation runs across two distinct traffic density setups on the same map topology:

Metric / Attribute	**Scenario 1: Standard Crosswalk**	**Scenario 2: High-Density Crowd**
Ego Spawn Coordinate	$(120.0, 132.0, 2.0)$	$(120.0, 132.0, 2.0)$

Metric / Attribute	Scenario 1: Standard Crosswalk	Scenario 2: High-Density Crowd
VRU Spawn (Base)	$(130.0, 141.0, 2.5)$	$(130.0, 141.0, 2.5)$
Total Pedestrian Count	6 Walkers	60 Walkers
Pedestrian Velocity	$4.2 \pm 0.4 \text{ m/s}$	$4.2 \pm 0.4 \text{ m/s}$
Baseline Speed Modifier	High-Speed Profile (Late Detection Testing)	High-Speed Profile (Late Detection Testing)

4. Key Performance Indicators (KPIs)

The simulation tracks and exports five distinct safety metrics to evaluate pipeline efficiency:

- **Collision Rate (collision):** Boolean indicating if the minimum distance dropped below the structural impact boundary ($< 1.95\text{m}$).
- **Minimum Distance to Target (min_dist):** The closest recorded distance vector between the vehicle bounding box and any active VRU agent.
- **Brake Trigger Timestamp (brake_time):** Chronological log in simulation seconds marking exactly when the vehicle changed its state out of cruise mode.
- **Trigger Velocity (trigger_speed):** Velocity profile entry speed (km/h) at the exact moment the braking command was issued.
- **Network Transport Latency (latency_ms):** Simulated communication lag applied exclusively to the V2X matrix (ranging randomly between 12.5ms and 18.2ms).

5. Summary of System Limitations / Code Review Notes

[!CAUTION]

Critical Edge Case Identified in Control Loop Loopback

During the main execution block loop, the script explicitly overrides the safety braking logic by systematically re-enabling the Traffic Manager Autopilot and resetting flags on **every single frame tick**:

Python

```
# Lines 188-192

ego_vehicle.set_autopilot(True)

autopilot_disabled = False
```

- **Impact:** This creates a race condition where the vehicle applies brakes for a fraction of a second, only to be immediately put back into cruise mode by the Traffic Manager on the subsequent tick. For valid data extraction, these lines should be wrapped in a logical block that executes *only after* the hazard has cleared.